

Privacy-Preserving Fingerprint Authentication

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Outline

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- 2 System Architecture & Feature Extraction
- 3 Quantization
- 4 Cryptography: The FLPSI Engine
- 5 Performance Evaluation

What is a Fingerprint?

Biological Identity

- Ridges & valleys form on the fingertip *in utero* and never change.
- **Unique:** even identical twins differ.
- **Immutable:** the pattern persists for life.
- **Universal:** everyone has ~ 10 fingers.
- These properties make it ideal ID material - but also **irreplaceable if leaked**.



Identity f : finger $\rightarrow$ pattern

$\Pr[\text{same pattern, different finger}] \approx 0$

How Fingerprint Authentication Works

The Standard Pipeline

- 1 **Enroll:** capture the finger, extract features, store a *template*.
- 2 **Authenticate:** capture again, extract features, compare to stored template.



$$\text{score}(e_{\text{probe}}, e_{\text{gal}}) > \theta \implies \text{genuine}$$

Here "features" are the 16-D CNN embedding and "compare" is the fuzzy PSI match - done in private.

The Problem

Biometrics are the worst password

- A password can be rotated. **A fingerprint cannot.**
- If a cryptographic key leaks - regenerate it. If a biometric leaks - **forever compromised.**

The trusting-honest-third-party catch

- Legacy methods authenticate via a private *matching* service - the matcher sees every biometric.
- **We need to verify a fingerprint without anyone learning it** - or even learning whether a match happened.

$$\underbrace{\text{verify}(x, Y)}_{\text{match-ness only}} \neq \underbrace{Y}_{\text{the database}}$$

Current Cryptographic Tools & Their Limits

Every prior family either **cannot tolerate biometric noise**, **loses accuracy**, or **pays a heavy cryptographic overhead**.

Tool	Citation	Approach	Limitation
One-way hash	standard	Compare digests $f_k(x)$	Exact match only
Fuzzy commitment	Juels & Wattenberg'99	XOR with an ECC codeword	Needs ordered features; leaks
Fuzzy vault	Juels & Sudan'02	Minutiae hidden in chaff	Correlation attacks; not revocable
Fuzzy extractors	Dodis et al.'04	Stable key + helper data	High error rate; entropy loss
Cancelable / BioHashing	Ratha'01; Teoh'04	Non-invertible keyed transform	Accuracy drop; token secrecy
HE (Blind-Touch)	Choi et al., AAAI'24	CKKS-encrypted matching	117 MB key; $\sim 5k$ cap; score only
FLPSI	Uzun'21; Bui & Cong'25	mOPRF + threshold PSI	Needs binary Hamming codes

The Closest Prior Work: Blind-Touch (AAAI 2024)

What it does

- A Siamese CNN maps a fingerprint to a **16-D embedding**; matching runs under CKKS homomorphic encryption, so the server never decrypts: $\text{Dec}(\text{Eval}(f, \text{Enc}(x))) = f(x)$.
- **We reuse its feature extractor** - the accuracy comes from the CNN, not the encryption.

Its three limitations - and our opening

- **Huge keys:** a 117 MB Galois rotation key on every server.
- **Scale ceiling:** $8,192 \text{ slots} \div 16 = 512 \text{ users per ciphertext}$; impractical beyond **$\sim 5,000$ users**.
- **No labels:** returns a match *score* only - never *which* record matched.

The matching layer is the bottleneck - so we replace it.

Our Proposed Solution: **Neural PSI**

What is Private Set Intersection (PSI)?

Formal Definition

- Two parties, sets X (Alice) and Y (Bob).
- They learn only the intersection $X \cap Y$ - and nothing else.

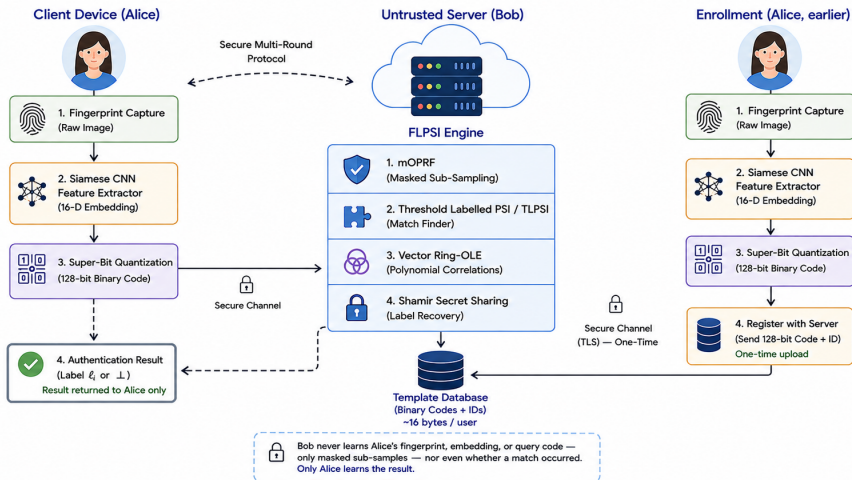
$$\text{PSI}(X, Y) \rightarrow X \cap Y, \quad \text{leakage } \mathcal{L} = X \cap Y \text{ only}$$

Why PSI for biometrics?

- Naive match: compare e to every $y \in Y$ - leaks mass information.
- **Asymmetric matching:** Alice learns the *label of her match*, Bob learns *nothing*.
- **But** biometric codes are *noisy* - exact set membership fails.
- $\Rightarrow$ **Fuzzy** / threshold PSI (FLPSI) tolerates t out of T sub-matches.

System Architecture

Neural PSI: Privacy-Preserving Fingerprint Authentication



Preprocessing & Convolution

Raw Input & Preprocessing

- 224×224 grayscale image normalized to $[0.0, 1.0]$.
- No alignment, filtering, or minutiae extraction.
- Augmented with rotation ($\pm 15^\circ$), translation ($\pm 6\%$), and Gaussian noise ($\sigma = 6$).

Five Convolutional Blocks

- $\text{Conv}(3 \times 3) \rightarrow \text{BatchNorm} \rightarrow \text{SiLU} \rightarrow \text{MaxPool}(2 \times 2)$.
- Channels grow $32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ while the map halves each block: $224 \rightarrow 112 \rightarrow 56 \rightarrow 28 \rightarrow 14 \rightarrow 7$.
- Progresses from detecting edges ($32 \times 112 \times 112$) to an abstract identity code ($512 \times 7 \times 7$).

The FC-16 Output

- Flattens the final spatial block ($512 \times 7 \times 7$) into **25,088** numbers.
- One linear layer $W \in \mathbb{R}^{25088 \times 16}$ produces the 16-D embedding $e \in \mathbb{R}^{16}$.
- **Client-side only:** The server never receives these continuous values.

Siamese Loss Function:

$$L = -\mathbb{E} \left[y \log \sigma(w \cdot d(e_1, e_2)) + (1 - y) \log (1 - \sigma(w \cdot d(e_1, e_2))) \right]$$

Why Super-Bit? Binarization Method Comparison

Three binarization schemes evaluated on SOCOFing - only Super-Bit achieves *both* similarity preservation and 50/50 bit balance.

	Method 1: $\text{sign}(> 0)$	Method 2: ITQ	Method 3: Super-Bit ★
Bits	16	16	128
Rule	$b_k = \mathbf{1}[e_k > 0]$	$b = \text{sign}((e - \mu)R)$	$b = \mathbf{1}[Pe_0 > \tau]$
Idea	Threshold each float at zero	Learn rotation R to align axes	Expand 16 floats $\rightarrow$ 128 bits
Issue	Borderline bits flip; only 2^{16} codes	Fixes flips, but still 16 bits	8 orthonormal blocks; median τ gives 50/50
EER 96²	10.86%	7.12%	5.14%
EER 224²	10.12%	5.55%	1.88%
Verdict	Rejected	Rejected	✓ Chosen

Only Super-Bit has enough code space *and* a minimum-variance angular estimate.

The Three Quantization Stages

Stage 1 - Population Centering

$$e_0 = e - \mu$$

Embeddings cluster in a few directions, so origin hyperplanes give **unbalanced bits**.
Subtract the population mean μ (computed offline).

Stage 2 - Super-Bit Projection

$$z = Pe_0, P^T = [Q_1, \dots, Q_8]$$

Random projections are redundant. Each block is Gaussian, then Gram-Schmidt orthonormalized ($Q_j^T Q_j = I_{16}$), giving a **minimum-variance** angular estimate.

Stage 3 - Balanced Median Thresholding

$$b_k = \mathbf{1}[z_k > \tau_k]$$

FLPSI requires **maximum entropy per bit** (50/50). The public median threshold τ guarantees it.

Evaluated and rejected - metric whitening. An optional $e_1 = \text{diag}(\sqrt{|w_1|})e_0$ step rescales axes by the head's learned weights. It helps a *weak* model (5.14% $\rightarrow$ 3.91%) but **hurts the trained one** (1.88% $\rightarrow$ 2.98%), so the deployed pipeline omits it.

Chosen Method: Super-Bit LSH

Super-Bit Projection & Thresholding

- Expands 16 floats into 128 bits using 8 orthonormal Gram-Schmidt blocks.
- Median τ guarantees a 50/50 balance, and it provides a minimum-variance angular estimate.

Projection Matrices & Binarization:

$$P^T = [Q_1, \dots, Q_8] \text{ with } Q_j^T Q_j = I_{16}$$

$$b_k = \mathbf{1}[z_k > \tau_k]$$

The Problem:

Two templates. Never exactly equal.

Two scans of the *same* finger almost never produce identical binary codes - noise, rotation, and pressure all shift a few bits.

What we actually need

$$d_H(x, y_i) \leq d_k \Rightarrow \text{reveal } \ell_i$$

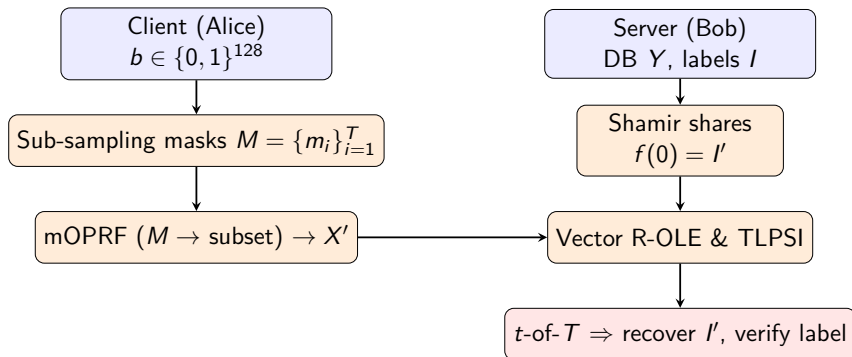
Ordinary PSI only tests *exact* equality ($x = y_i$). It has no notion of “close enough.”

Enter FLPSI

Fuzzy Labelled Private Set Intersection tolerates bounded Hamming error under a precise leakage profile $\mathcal{L}$:

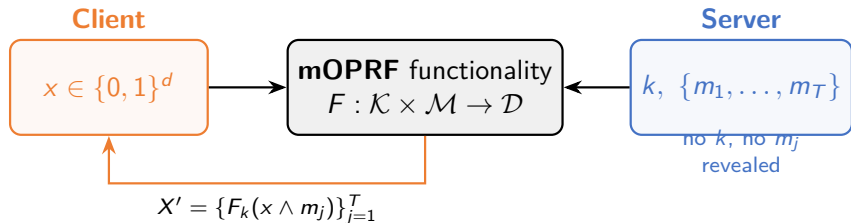
- Server learns **nothing**.
- Client learns the **label** on a match, *plus* which sub-samples collided - never the stored template itself.

Cryptography Stack



Primitive 1: Masked Oblivious PRF (mOPRF)

The problem: sub-sampling $x \wedge m_j$ directly would leak biometric bits whenever a match occurs.



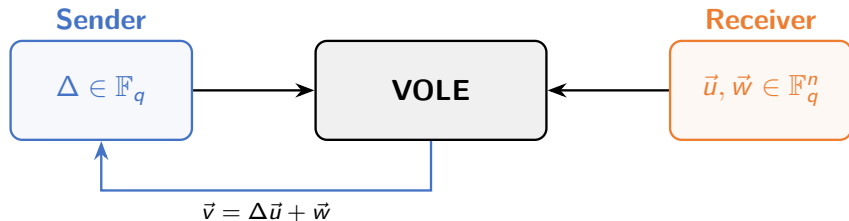
Why it works

$$F_k(x \wedge m_j) = F_k(y \wedge m_j) \iff x \wedge m_j = y \wedge m_j$$

Equality of sub-samples is preserved; the sub-samples themselves are computationally hidden - deterministic *and* pseudorandom.

Primitive 2: Vector Oblivious Linear Evaluation (VOLE)

The problem: generating n independent correlations via n **Oblivious Linear Evaluation (OLE)** calls is far too slow for millions of matches.



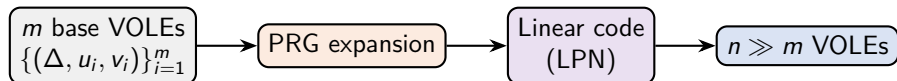
The linear correlation

$$v_i = u_i \Delta + w_i, \quad 1 \leq i \leq n \quad \Longleftrightarrow \quad \vec{v} = \Delta \vec{u} + \vec{w}$$

Sender knows $(\Delta, \vec{v})$; **receiver** knows $(\vec{u}, \vec{w})$ - neither alone can reconstruct the relation, yet together it holds exactly. This one relation is the seed for everything that follows.

Silent VOLE Extension: From a Few to Millions

The problem: FLPSI needs *millions* of VOLE correlations per query - one base VOLE per correlation is impossibly expensive.



Why LPN?

- Expands a few VOLEs into millions.
- Preserves VOLE correlations.
- Computationally indistinguishable from random.

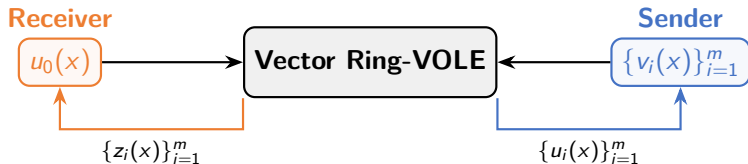
Primitive 3: Vector Ring-VOLE - VOLE Meets Polynomials

The problem: TLPSI needs *polynomial products*, not just scalar correlations - Vector-VOLE alone cannot multiply polynomials.

Setup (evaluation-point trick)

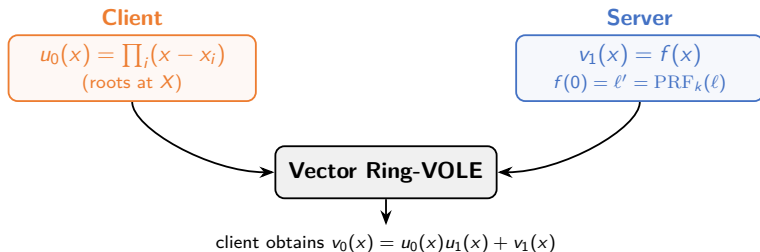
Receiver holds $u_0(x)$, $\deg \leq d$; sender holds $\{v_i(x)\}_{i=1}^m$, $\deg \leq 2d$. Both evaluate over a public set $X = \{x_1, \dots, x_{2d+1}\}$:

$$U_i = u_i(X), \quad V_i = v_i(X), \quad Z_i = u_0(X) \odot U_i + V_i$$



From Ring-VOLE to Threshold Labelled PSI (TLPSI)

The idea: represent a *set* by the polynomial whose roots are its elements.

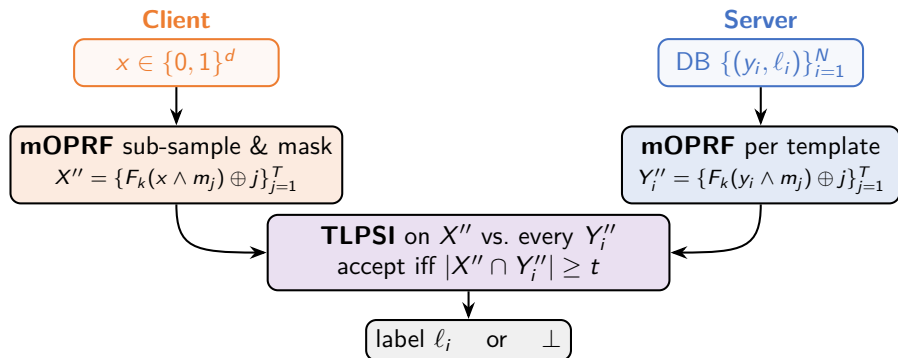


The magic identity

For $x_i \in X \cap Y$: $u_0(x_i) = 0 \Rightarrow v_0(x_i) = v_1(x_i) = f(x_i)$.

Every common element yields one true evaluation of the Shamir polynomial f . Collect $\geq t$ of them $\Rightarrow$ interpolate $f(0) = \ell'$; reveal ℓ after checking validity. Fewer than $t \Rightarrow$ information-theoretically hidden.

The Full FLPSI Protocol: mOPRF + TLPSI



Intuition: Closeness Translates to Threshold Intersection

If the Hamming distance between the templates is small, enough sub-samples will naturally collide to hit our intersection threshold t .

Theorem (informal)

NeuralPSI securely realizes FLPSI in the $\{\mathcal{F}_{\text{TLPSI}}^{T,t}, \mathcal{F}_{\text{mOPRF}}^T\}$ -hybrid model against semi-honest adversaries.

- 1 CNN runs strictly locally - no raw biometric ever leaves the client.
- 2 Server only ever sees **PRF-masked sub-samples** of $\tilde{x}$, never $\tilde{x}$ itself.
- 3 The core FLPSI argument composes securely via a standard hybrid argument.

Demonstration

Final Results & Accuracy

Accuracy & Storage Milestones

- Retains 1.88% EER, comparable to continuous float accuracy ceilings.
- Achieves a 107x reduction in per-user template size (down to 16 Bytes).

Table 1: Open-set EER on SOCOFing (224 × 224 CNN model)

Representation	Bits	EER
Float features (HE ceiling)	-	0.33%
sign(> 0)	16	10.12%
ITQ	16	5.55%
Super-Bit (recommended)	128	1.88%
+ metric whitening	128	2.98%

Latency, Storage & Communication

	Blind-Touch (HE/CKKS)	NeuralPSI (ours)
Servers used	1 main + 3 cluster	1
Latency, single server	1,334 ms	~824 ms
Template per user	1.67 KB	16 B
Evaluation (Galois) key	117 MB	none
Per-query data ($N = 5,000$)	856 KB ciphertext	7.77 MB
per record	-	1.54 KB
at $10^5 / 10^6$	-	~150 MB / ~1.5 GB
Capacity	512 users/ciphertext	linear in N

Blind-Touch: Choi, Woo & Kim, AAAI 2024 (Tables 2 and 8). NeuralPSI measured single-host on the real protocol binary: 0.80 ms CNN + 823 ms matching; $10^5/10^6$ extrapolated from $\approx 0.25 \text{ MB} + 1.54 \text{ KB} \cdot N$, against Bui & Cong's reported 153 MB / 1533 MB.

Conclusion & Future Work

Conclusion

- Fingerprint authentication **without revealing the biometric**, no trusted third party.
- The **binarization bridge** makes a CNN and Fuzzy Labelled PSI compatible.
- HE $\rightarrow$ VOLE + Shamir: **16 B/user** ($\sim 107\times$ smaller), **no 117 MB key**; **sub-second** at $N = 5,000$, and linear + shardable past 10^6 .

Future Work

- **Larger datasets:** cross-sensor / cross-session capture - the real credibility test.
- **Algorithm:** accuracy saturates at 128 bits, so the **16-D embedding** is the bottleneck - widen it; extend to malicious security.
- **Parallel processing:** the FLPSI scan is linear and embarrassingly parallel - shard it, precompute the offline phase.
- **Research paper:** formal security proof + integrated prototype.

Thank You

Questions?